24. Topological code verification and noise-specific decoder evaluation

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Imagine storing information in loops spread across a lattice so that a small local disturbance usually cannot change the stored value. A topological quantum code uses a quantum version of this idea. Local measurements reveal the endpoints of error patterns, and a decoder tries to join those endpoints correctly. The subtlety is that two repairs can have the same endpoints yet differ by a loop that changes the logical information.

Key terms

- **Toric code** — A quantum error-correcting code on a lattice with periodic boundaries, topologically equivalent to a torus.
- **Surface code** — A related geometric code with carefully chosen open boundaries.
- **Logical operator** — An operation changing the encoded information while remaining consistent with the code's checks.
- **Noncontractible loop** — A loop winding around the torus that cannot be shrunk to a point within it.
- **Homology** — A mathematical way of distinguishing cycles modulo boundaries; here it identifies whether a closed error pattern acts logically.
- **Syndrome** — Check outcomes locating the boundary of an error chain.
- **Recovery** — An operation chosen to cancel the error; matching the syndrome is necessary but does not ensure logical success.
- **Minimum-weight matching** — A graph optimization method used in certain decoders to pair defects efficiently.
- **Logical failure probability** — The probability that error and recovery together change encoded information.
- **Threshold** — An asymptotic noise boundary for successful suppression of logical errors in a specified code-and-decoder family.

Why this matters

Geometric codes connect an intuitive picture of local errors to precise algebraic guarantees. They are important models for protecting quantum information using structured measurements. Understanding logical failure, rather than only syndrome cancellation, is essential for assessing whether increasing the code size actually improves reliability.

What progress would mean

A useful result would prove the code's parameters and measure a fixed decoder under a declared noise model. An estimated threshold would remain specific to that model and decoder, not a universal device error tolerance.

Core question

Can a toric-code family's algebraic parameters be proved, and can a fixed decoder's logical failure probability be evaluated under a precisely specified error model?

Scope and assumptions

Use an L×L square torus with qubits on edges and L≥3. Start with independent Z errors of probability p and perfect syndrome extraction. Specify the decoder, tie-breaking rule, and whether recovery weights are uniform or likelihood based.

Keep exact code properties separate from statistical decoder performance. Planar boundaries, color codes, measurement noise, and circuit-level noise are distinct extensions.

Mathematical target

Over F_2 use $C_2 \to \partial_2$ $C_1 \to \partial_1$ C_0 with $\partial_1\partial_2=0$, $H_X=\partial_1$, $H_Z={\partial_2}^T$. Then $n=2L^2$, $k=2$, and prove $d=L$ using noncontractible primal and dual cycles.

For Z error e and recovery r, success means $e+r \in \mathrm{im}\ \partial_2$, not merely $\partial_1(e+r)=0$. A threshold assertion concerns the limit of $P_{fail}(L,p)$ as $L\to\infty$ for this fixed noise/decoder family; a finite crossing is an estimate.

Required outputs and proof obligations

Export lattice incidence data, ranks and logical bases, distance argument, decoder specification, and failure counts with confidence intervals. Exhaustive small-size evaluation can give an exact failure polynomial in p. Monte Carlo output must retain trials and failure counts, including zero-failure upper intervals.

Validation and rejection controls

Check n,k,d on several small tori. Verify logical anticommutation via primal/dual intersection. Enumerate all low-weight errors on tiny instances and test the stabilizer-membership success criterion. Validate a matching solution's optimality separately from logical decoding success.

Milestones and research extension

Baseline: exact construction and distance proof with a tested decoder. Strong: reproducible finite-size failure curves and uncertainty-aware threshold estimates. Research extension: a rigorous threshold bound or noisy-measurement/circuit model, with no transfer of numerical thresholds across noise conventions.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Dennis, Kitaev, Landahl and Preskill, Topological quantum memory — Homological coding and model-dependent error-correction thresholds.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0, \quad \partial_1\partial_2 = 0, \quad H_X = \partial_1, \quad H_Z = \partial_2^{\mathsf{T}},$$

$$n = 2L^2, \quad k = 2, \quad d = L \quad (L \geq 3), \qquad e + r \in \operatorname{im}\partial_2 \quad \text{for successful } Z\text{-error recovery.}$$

A vanishing residual syndrome only gives $e + r \in \ker \partial_1$ and can hide a nontrivial logical cycle. Statistical decoder curves require the fixed noise model and decoder; finite-size crossings are threshold estimates, not limit theorems.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — noise mismatch (17, 88, 1003).** A ~10.9% figure is not a universal depolarizing/circuit threshold. Noise channel, measurement assumptions, and decoder determine the comparison.
2. **Major — coefficient field (97–107).** CSS homology is computed over F_2; writing $\mathrm{H}_1{=}\mathrm{Z}^2$ without identifying the coefficient change confuses topological and binary-code data.
3. **Blocking — syndrome versus correction (110–113, 133–136).** Two chains with the same boundary may differ by a logical loop. Success requires trivial homology of error plus recovery.

4. **Major — certification boundary (49–52, 136).** Monte Carlo threshold fits are not algebraic proofs, and prohibiting numerical optimization conflicts with matching and simulation plans. The revision allows useful numerical work while reserving proof language for justified conclusions.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **Can a toric-code family's algebraic parameters be proved, and can a fixed decoder's logical failure probability be evaluated under a precisely specified error model?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: exact construction and distance proof with a tested decoder. Strong: reproducible finite-size failure curves and uncertainty-aware threshold estimates. Research extension: a rigorous threshold bound or noisy-measurement/circuit model, with no transfer of numerical thresholds across noise conventions.

Sources supporting the corrections

- Dennis, Kitaev, Landahl and Preskill, Topological quantum memory — Homological coding and model-dependent error-correction thresholds.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.

- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.